Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

Aim: **Image Enhancement Using Smoothing and Sharpening Techniques**

Theory:
Smoothing and sharpening are common image enhancement techniques used to modify the spatial characteristics of an image.

Smoothing:

- Smoothing, also known as blurring, is used to reduce noise and remove fine details in an image.
- It works by applying a low-pass filter that averages the pixel values within a neighborhood, thus producing a blurred effect.

Sharpening:

- Sharpening is used to enhance the edges and fine details in an image.
- It works by applying a high-pass filter that amplifies the differences between neighboring pixel values, thus increasing the contrast and emphasizing edges**.**

MEAN FILTER

The mean filter replaces each pixel by the average of the pixels in the neighborhood.
It is a simple low pass filter and is useful for reducing random noise.

For a 3x3 mean filter the coefficient is 1/9 and the kernel is:
[1 1 1]
[1 1 1]
[1 1 1]

It is computationally inexpensive and simple to implement useful for noise reduction.

WEIGHTED MEAN FILTER

The weighted mean filter is similar to mean filter but neighboring pixels do not need to contribute equally.
The kernel coefficients are normalized by their total sum so that overall brightness is maintained.

For ex:
[1 1 2]
[1 3 1]
[1 1 7]
Coefficient = 1/18

	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

It reduces small intensity variations and can preserve local information better than uniform averaging.

GAUSSIAN FILTER

It is a smoothing technique in which pixels near the centre of the neighbourhood receive higher weights than pixels farther away.
The weights follow the general shape of a Gaussian distribution.

For ex:
Kernel =
[1 2 1]
[2 4 2]
[1 2 1]
Coefficient = 1/16

This filter gives smooth noise reduction and generally produces a natural looking blur.

LAPLACIAN FILTER

Laplacian is a second order derivative operator used to detect regions where the intensity changes rapidly.
It is used for edge detection and image sharpening.

For ex:

Kernel=
[0 1 0]
[1 -4 1]
[0 1 0]

The Laplacian image highlight locations of rapid intensity change.

LAPLACIAN SHARPENING

It uses Laplacian component to enhance the original image.
Depending upon the sign convention of the Laplacian kernel the sharpening operation is written.
The sharpened image has more prominent edges and local details than the original.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
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Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def mean_filter(image):
    mean_3x3 = cv2.blur(image, (3,3))
    mean_5x5 = cv2.blur(image, (5,5))

    return mean_3x3, mean_5x5

def weighted_mean_filter(image):

    weighted_kernel = np.array([
        [1,1,2],
        [1,3,1],
        [1,1,7]
    ], dtype=np.float32)

    weighted_kernel = weighted_kernel / 18

    weighted_mean = cv2.filter2D(
        image,
        -1,
        weighted_kernel
    )

    return weighted_mean

def gaussian_filter(image):

    gaussian_kernel = np.array([
        [1,2,1],
        [2,4,2],
        [1,2,1]
    ], dtype=np.float32)

    gaussian_kernel = gaussian_kernel / 16

    gaussian_mean = cv2.filter2D(
        image,
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
        -1,
        gaussian_kernel
    )

    return gaussian_mean

def laplacian_sharpening(image):

    laplacian = cv2.Laplacian(
        image,
        cv2.CV_32F
    )

    laplacian_abs = cv2.convertScaleAbs(
        laplacian
    )

    sharpened = cv2.convertScaleAbs(
        image.astype(np.float32) - laplacian
    )

    return laplacian_abs, sharpened
```

IMAGE – 1

Code:

```
image1 = cv2.imread("images/IMAGE1.png")

image1_rgb = cv2.cvtColor(
    image1,
    cv2.COLOR_BGR2RGB
)

mean1_3x3, mean1_5x5 = mean_filter(image1)

weighted1 = weighted_mean_filter(image1)

gaussian1 = gaussian_filter(image1)

laplacian1, sharpened1 = laplacian_sharpening(image1)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image1_rgb)
plt.title("Original Image 1")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(
    cv2.cvtColor(mean1_3x3, cv2.COLOR_BGR2RGB)
)
plt.title("Mean Filter 3x3")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(
    cv2.cvtColor(mean1_5x5, cv2.COLOR_BGR2RGB)
)
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(
    cv2.cvtColor(weighted1, cv2.COLOR_BGR2RGB)
)
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(
    cv2.cvtColor(gaussian1, cv2.COLOR_BGR2RGB)
)
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
plt.imshow(image1_rgb)
plt.title("Original Image 1")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(laplacian1, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

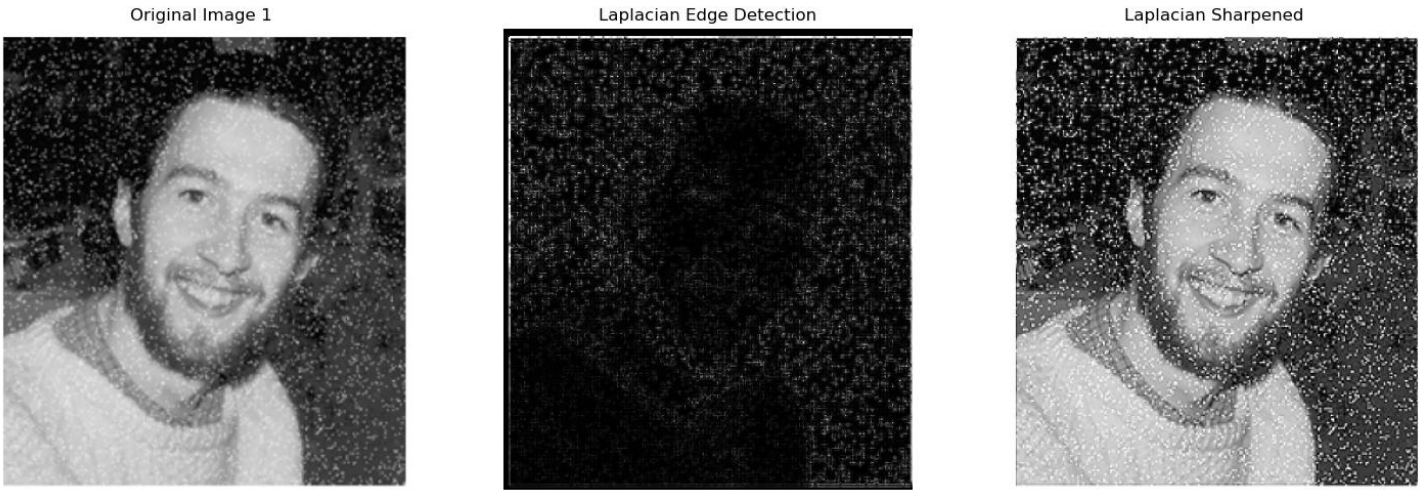
plt.subplot(1,3,3)
plt.imshow(
    cv2.cvtColor(sharpened1, cv2.COLOR_BGR2RGB)
)
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**



OBSERVATION

- Mean 3×3 reduces the noise slightly while preserving few details.
- Mean 5×5 provides stronger smoothing but makes fine details softer.
- Weighted Mean and Gaussian filters provide moderate smoothing with better detail preservation.
- Laplacian edge detection highlights facial edges as well as noisy regions.
- Laplacian sharpening makes the edges more prominent but also increases the visible noise.

CONCLUSION

Gaussian / Weighted Mean give a good balance between noise reduction and detail preservation.

IMAGE – 2
Code:

```
image2 = cv2.imread("images/IMAGE2.png")

image2_rgb = cv2.cvtColor(
    image2,
    cv2.COLOR_BGR2RGB
)

mean2_3x3, mean2_5x5 = mean_filter(image2)

weighted2 = weighted_mean_filter(image2)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
gaussian2 = gaussian_filter(image2)

laplacian2, sharpened2 = laplacian_sharpening(image2)

plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image2_rgb)
plt.title("Original Image 2")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(cv2.cvtColor(mean2_3x3, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 3x3")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(cv2.cvtColor(mean2_5x5, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(cv2.cvtColor(weighted2, cv2.COLOR_BGR2RGB))
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(gaussian2, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image2_rgb)
plt.title("Original Image 2")
plt.axis("off")
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
plt.subplot(1,3,2)
plt.imshow(laplacian2, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(cv2.cvtColor(sharpened2, cv2.COLOR_BGR2RGB))
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:





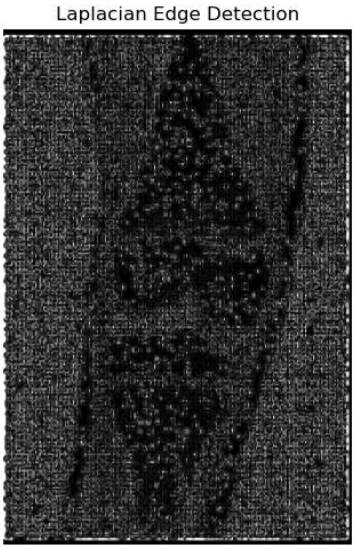






Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**







OBSERVATION

- All the smoothing filters reduce the granular noise to some extent.
- Mean 3x3 provides mild smoothing
- Mean 5x5 gives the strongest blur.
- Weighted Mean and Gaussian filters smooth the image.
- Laplacian highlights boundaries along with noise
- Laplacian sharpening makes the boundaries more prominent.

CONCLUSION

Smoothing improves overall appearance of noisy image while sharpening improves structural details

IMAGE – 3
Code:

```
image3 = cv2.imread("images/IMAGE3.png")

image3_rgb = cv2.cvtColor(
    image3,
    cv2.COLOR_BGR2RGB
)

mean3_3x3, mean3_5x5 = mean_filter(image3)

weighted3 = weighted_mean_filter(image3)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
gaussian3 = gaussian_filter(image3)

laplacian3, sharpened3 = laplacian_sharpening(image3)

plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image3_rgb)
plt.title("Original Image 3")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(cv2.cvtColor(mean3_3x3, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 3x3")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(cv2.cvtColor(mean3_5x5, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(cv2.cvtColor(weighted3, cv2.COLOR_BGR2RGB))
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(gaussian3, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image3_rgb)
plt.title("Original Image 3")
plt.axis("off")
```

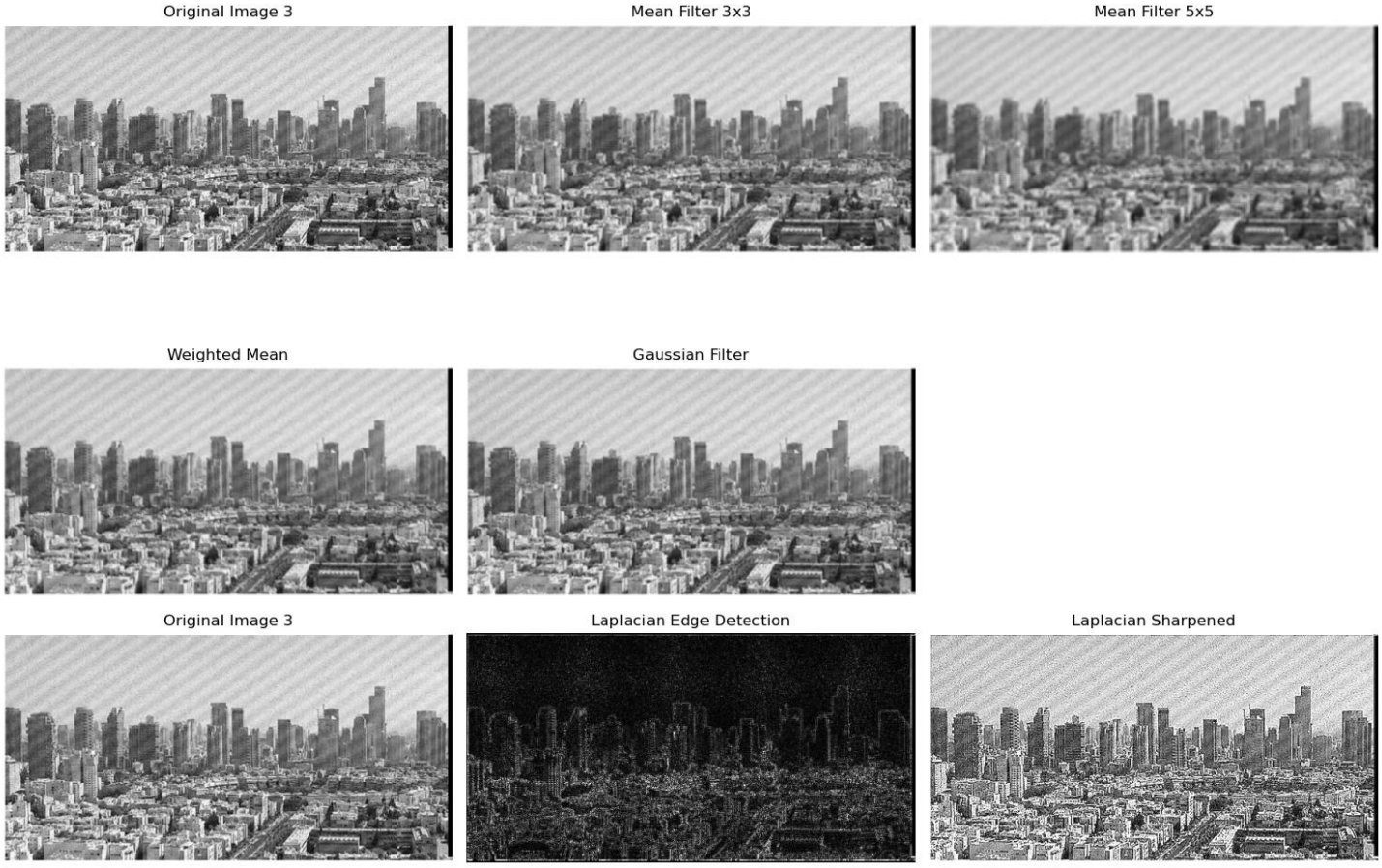
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
plt.subplot(1,3,2)
plt.imshow(laplacian3, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(cv2.cvtColor(sharpened3, cv2.COLOR_BGR2RGB))
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

OBSERVATION

- Mean 3x3 gives slight smoothing while maintain building details.
- Mean 5x5 gives stronger smoothing and makes edges softer.
- Weighted Mean and Gaussian provide smoother transitions while main structure is preserved.
- Laplacian edge detection highlights boundaries of building.
- Laplacian sharpening makes edges and details clear.

CONCLUSION

Moderate smoothing preserves important details while sharpening improves visibility of edges.

IMAGE – 4

Code:

```
image4 = cv2.imread("images/IMAGE4.png")

image4_rgb = cv2.cvtColor(
    image4,
    cv2.COLOR_BGR2RGB
)

mean4_3x3, mean4_5x5 = mean_filter(image4)

weighted4 = weighted_mean_filter(image4)

gaussian4 = gaussian_filter(image4)

laplacian4, sharpened4 = laplacian_sharpening(image4)

plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image4_rgb)
plt.title("Original Image 4")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(cv2.cvtColor(mean4_3x3, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 3x3")
```

```
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(cv2.cvtColor(mean4_5x5, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(cv2.cvtColor(weighted4, cv2.COLOR_BGR2RGB))
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(gaussian4, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image4_rgb)
plt.title("Original Image 4")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(laplacian4, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(cv2.cvtColor(sharpened4, cv2.COLOR_BGR2RGB))
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

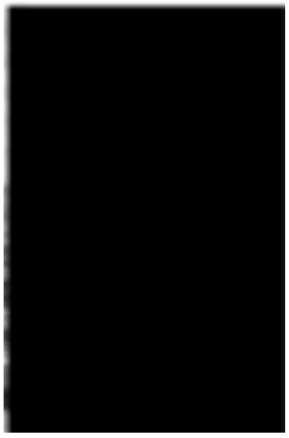
Original Image 4



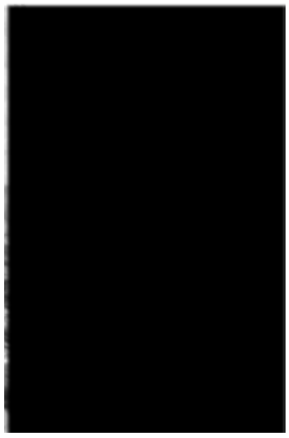
Mean Filter 3x3



Mean Filter 5x5



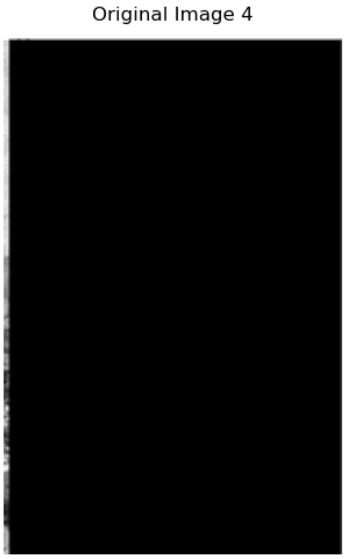
Weighted Mean

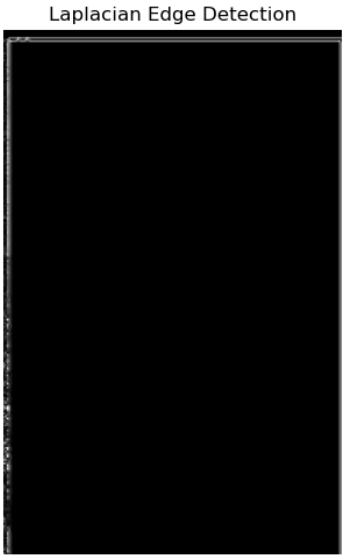


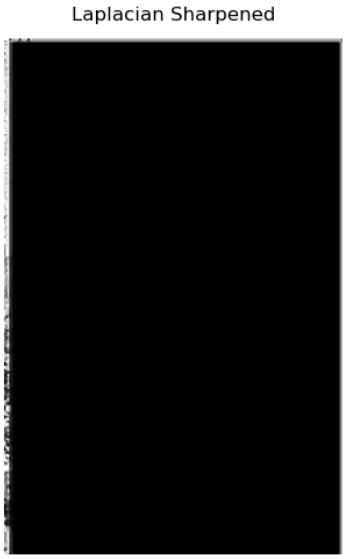
Gaussian Filter



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**







OBSERVATION

- The input image is extremely dark.
- Mean 3x3 and Mean 5x5 produce no change.
- Weighted Mean and Gaussian filters also show no visible improvement.
- Laplacian edge detection does not enhance and edges or boundaries.
- Laplacian sharpening also does not produce significant improvement.

CONCLUSION

There is no improvement in the image because of its very dark appearance.

IMAGE – 5
Code:

```
image5 = cv2.imread("images/IMAGE5.png")

image5_rgb = cv2.cvtColor(
    image5,
    cv2.COLOR_BGR2RGB
)

mean5_3x3, mean5_5x5 = mean_filter(image5)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

```
weighted5 = weighted_mean_filter(image5)

gaussian5 = gaussian_filter(image5)

laplacian5, sharpened5 = laplacian_sharpening(image5)

plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image5_rgb)
plt.title("Original Image 5")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(cv2.cvtColor(mean5_3x3, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 3x3")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(cv2.cvtColor(mean5_5x5, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(cv2.cvtColor(weighted5, cv2.COLOR_BGR2RGB))
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(gaussian5, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image5_rgb)
plt.title("Original Image 5")
```

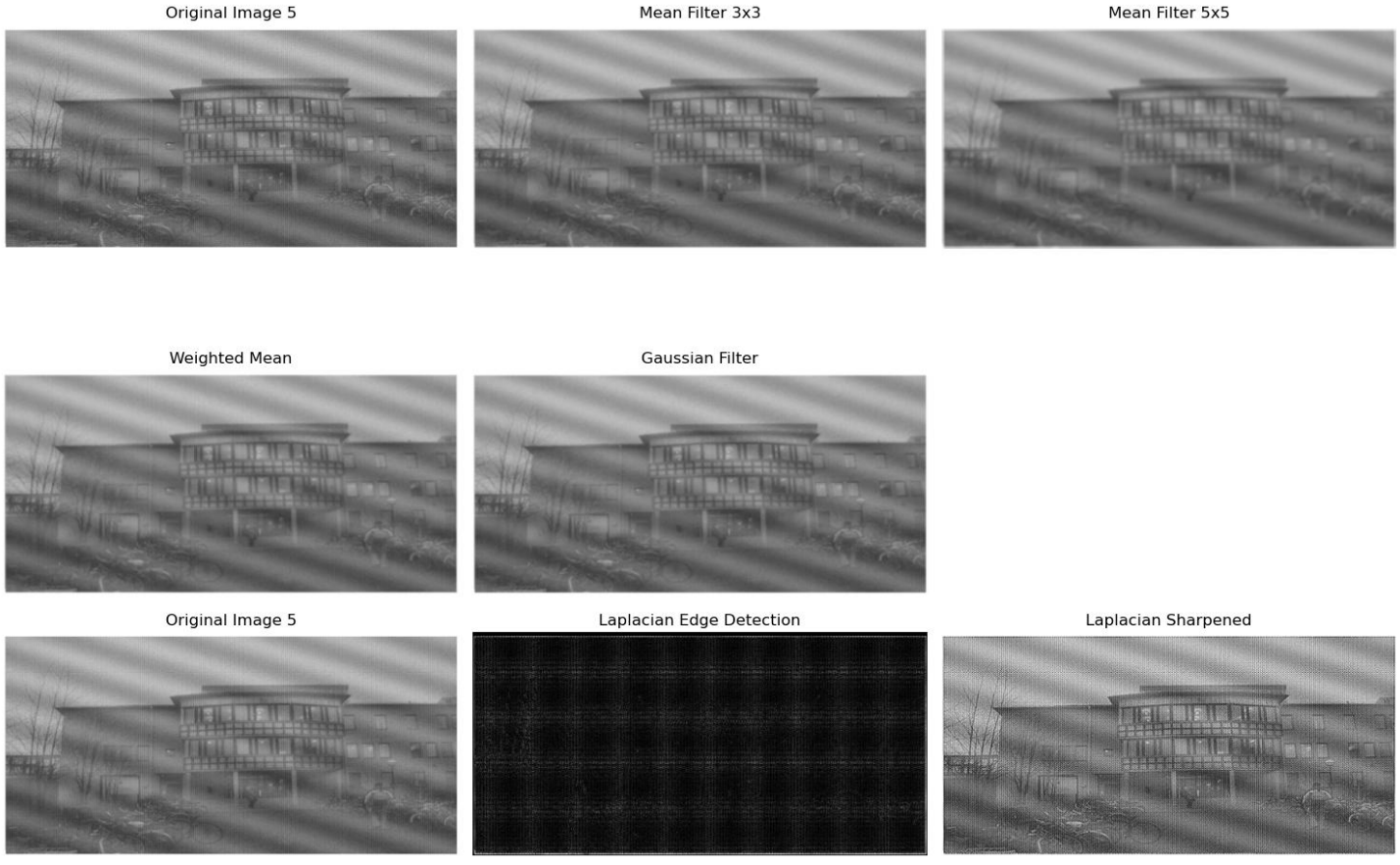
```
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(laplacian5, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(cv2.cvtColor(sharpened5, cv2.COLOR_BGR2RGB))
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**

OBSERVATION

- Mean 3x3 provides slight smoothing
- Mean 5x5 produces the strongest blur and makes the building even softer.
- Gaussian and Weighted Mean preserve the main structure better.
- Laplacian edge detection highlights building edges.
- Laplacian sharpening enhances the structural edges of the building.

CONCLUSION

Gaussian / Weighted Mean provide suitable smoothing while sharpening improves visibility of structures.

IMAGE – 6
Code:

```
image6 = cv2.imread("images/IMAGE6.png")

image6_rgb = cv2.cvtColor(
    image6,
    cv2.COLOR_BGR2RGB
)

mean6_3x3, mean6_5x5 = mean_filter(image6)

weighted6 = weighted_mean_filter(image6)

gaussian6 = gaussian_filter(image6)

laplacian6, sharpened6 = laplacian_sharpening(image6)

plt.figure(figsize=(15,8))

plt.subplot(2,3,1)
plt.imshow(image6_rgb)
plt.title("Original Image 6")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(cv2.cvtColor(mean6_3x3, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 3x3")
plt.axis("off")
```

```
plt.subplot(2,3,3)
plt.imshow(cv2.cvtColor(mean6_5x5, cv2.COLOR_BGR2RGB))
plt.title("Mean Filter 5x5")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(cv2.cvtColor(weighted6, cv2.COLOR_BGR2RGB))
plt.title("Weighted Mean")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(gaussian6, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Filter")
plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(image6_rgb)
plt.title("Original Image 6")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(laplacian6, cmap="gray")
plt.title("Laplacian Edge Detection")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(cv2.cvtColor(sharpened6, cv2.COLOR_BGR2RGB))
plt.title("Laplacian Sharpened")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Image Enhancement Using Smoothing and Sharpening Techniques**	
Open-ended-Problem: 09	**Date:06-09-2026**	**Enrollment No:92400133121**



OBSERVATION

- Mean 3x3 filter produces additional smoothing on given input blurred image.

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- Mean 5x5 filter produces stronger smoothing and makes the image even softer which reduces the visibility of fine details.
- Weighted Mean and Gaussian filters also smooth existing blurred regions
- Laplacian edge detection shows some edges.
- Laplacian sharpening slightly enhances available edges but cannot recover lost details.

CONCLUSION

Comparatively sharpening is better for this image as at least some of the existing edges are enhanced while smoothing makes the image even blur/softer.

Overall Conclusion

- Smoothing reduces noise and unwanted variations but may cause loss of some details.
- Sharpening enhances edges and improves clarity but can amplify noise.
- According to the type of the image, we need to choose appropriate filter.